


Economics and Business Administration

# Business Organizations and Organizational structures

Prof. Leonardo Boni

# Main concepts



- What is the organization
- The concept of organizational structure
- Foundations of organizational structure
- Decisions in organizational design
    - **microstructure:** positions and roles of people involved
    - **macrostructure**: organizational structures

# The organization and growth dynamics



**Organization**: A group of individuals with a common goal, pursued through appropriately defined and coordinated individual and collective actions.

As an organization grows, its complexity increases because

- the number of activities to be carried out
- the diversity of activities to be carried out
- the number of members of the organization

**There could be problems of:**

- • Information: Not everyone has the same information set, information gets lost/distorted
- • Goal alignment: Not everyone pursues organizational goals, but also pursues personal goals
- • Coordination: Aligning information, actions, and goals requires time and resources (coordination costs arise)

# Division of labour – Adam Smith (1780 circa)

• The development of nations depends on the organization and structuring of businesses in relation to market dynamics.

• The ability to organize **individual skills** around projects of collective interest. **First, for the business, and then for the nation.**

• Two fundamental concepts in the organization of work.

**Specialization**: the entrepreneur grants employees the freedom to express their skills.

**Complementarity**: joint specializations identify a collective production process.

This mix **generates economies of scale and learning, evolving the potential of labor.**

# Evolutions of the thoughts – Marshall (1890) and Taylor (1910)

- **Marshall:** Industrial and manufacturing development led to the evolution of thinking about the division of labor.

- **The unit of analysis changed: from firm to territory.**
- **Industrial districts introduced the concept of complementarity and specialization between firms, not within firms.**

**Loss of the individual vision of specialization and corporate organization, resulting in the generation of mass production models.**

- **Taylor:** Totally rational and scientific organization of work, viewing the firm as a perfect and aseptic system, the only way to organize work.
- Division of labor was aseptic and independent of skill development.
    - **Adoption of the Taylorist model à la Ford, 1915**

**"Any customer can have his car painted any color he wants, as long as it's black" Henry Ford, about Model T.**



Fordism builds on Taylorist assumptions, adding the concept of mass production and standardization.

# Evolution of thoughts - Theory of growth of organizations – Edith Penrose (1950 – )

## *III.a. The Penrosean Firm and the Market*

The firm in *TGF* is a collection of productive resources (human and non-human) under administrative coordination and authoritative communication that produces goods and services for sale in the market for a profit (Penrose 1959, 1985, 1995). Administrative coordination and 'authoritative communication' define the boundaries of the firm. Very much like Coase (1937), but without having at the time been influenced by his classic article, Penrose maintained the distinction between the firm and the market.

The essential difference between economic activity inside the firm and economic activity in the 'market' is that the former is carried on within an administrative organization, while the latter is not (1995, p. 15).[12]

The boundary of the firm is what distinguishes it from the market and therefore must 'exist' whether or not it is 'real' . . . (p. xvi).[13]

# Evolution of thoughts – Theory of growth of organizations – Edith Penrose (1950 – )

Edith Penrose key pillars: **RESOURCES + GENERATION OF SERVICES**

- Resource-based view:

Penrose theory emphasizes that firms are essentially **bundles of resources, and that the services provided by those resources are what define a firm's uniqueness.**

- Managerial competences:

The ability of **managers to effectively employ, combine, and develop resources** is fundamental to the success of a business.

- Services as growth enablers:

The **services provided by resources, not the resources themselves**, that drive a company's operations and growth.

- Dynamic interaction:

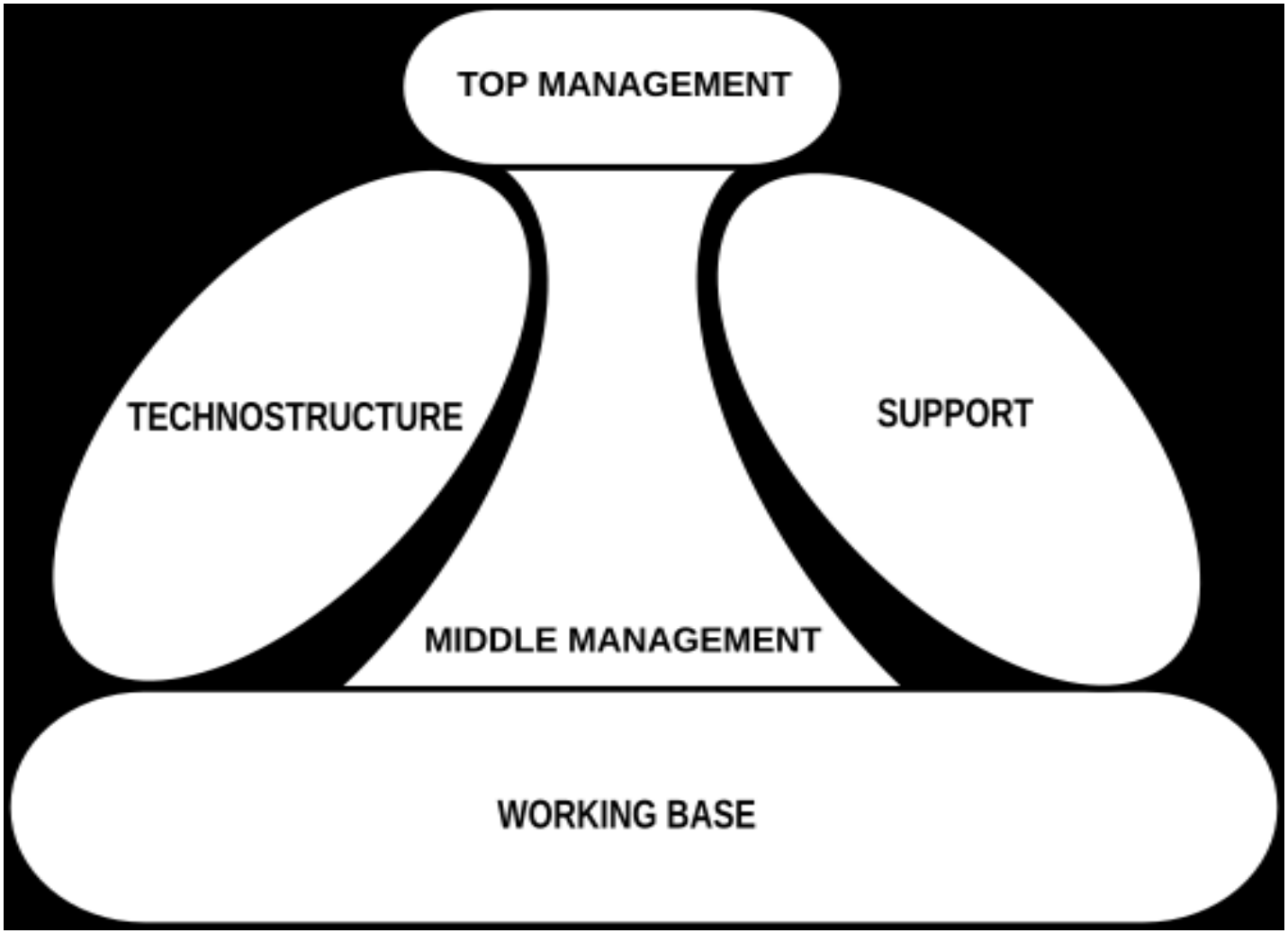
**Firms are viewed as entities that constantly interact with their environment, adapting the use of resources to create new services and opportunities.**

- Endogenous growth:

Business growth is determined by the company's internal capabilities and resources, rather than solely by external factors.

# Parts of the organizations: Mintzberg model





# Organizational design

Organizational design aims to find the best ways to coordinate activities in which interdependent organizational actors employ scarce resources to achieve valuable goals.

Typically, organizations are designed by the organizational top management (the "bosses").

- ***shareholders***, whose power derives from the ownership
- Those to whom the shareholders have appointed the administration of the firm (es: chief executive officers)

# Organizational design

The result of the organizational design:

- **Organizational structure:** represents the set of authority (hierarchical) and communication channels that are established between the different roles and organizational units.

- **Microstructure design** → Individual focus
  - define the position and roles of individuals in the organization, i.e. what they do and how they should do it

- **Macrostructure design** → Organizational unit
  - decide how to allocate activities among the various organizational units (groups) and define the overall organizational structure

# Job

**Microstructure**

- **Job**: set of tasks characterizing an individual's position, i.e., an individual's position within the organization
  - **Task**: set of connected and inseparable actions
  - **Individual position**: position of the individual within the organization to which a job is associated
  - **Role**: expectations regarding the behavior expected of those who hold a given individual position

- Fundamental aspects
  1. **Specialization and design of coordination mechanisms**
  2. **Formalization of the content and methods of what individuals do within the organization**
  3. **Decision-making autonomy and design of incentives to align**
  4. **individual objectives with organizational ones**
  5. **Development of individual skills/competencies and motivation management**

# Specialization



**Horizontal specialization:** refers to the way (different) tasks are assigned among multiple individuals; coordinating individuals performing various tasks.

**Vertical specialization:** separation between planning/control and task execution.

- Distinction (not always rigid) between manager (who decides what needs to be done and ensures it is done) and maker (who executes)
- Introduces a hierarchy among individuals' positions in the organization

# Specialization: pros and cons



**Pros**

Specialization reduces costs and increases efficiency as:

- Promotes learning
- Reduces training costs
- Reduces downtime
- Standardizes work methods

**Cons**

- Alienation
- Loss of expertise
- Increased coordination costs

# Coordination mechanisms

**Microstructure**

Coordination costs can be reduced by adopting appropriate coordination mechanisms.

Coordination mechanisms can be:

- **Formal**, such as organizing a meeting every Thursday among team members to coordinate future actions/decisions.
- **Informal**, for example, two SJs have known each other for a long time and "understand each other immediately."

We focus on the 5 fundamental coordination mechanisms, which have an increasing level of complexity

1. **Mutual adaptation (mutual adjustment)**
2. **Direct supervision**

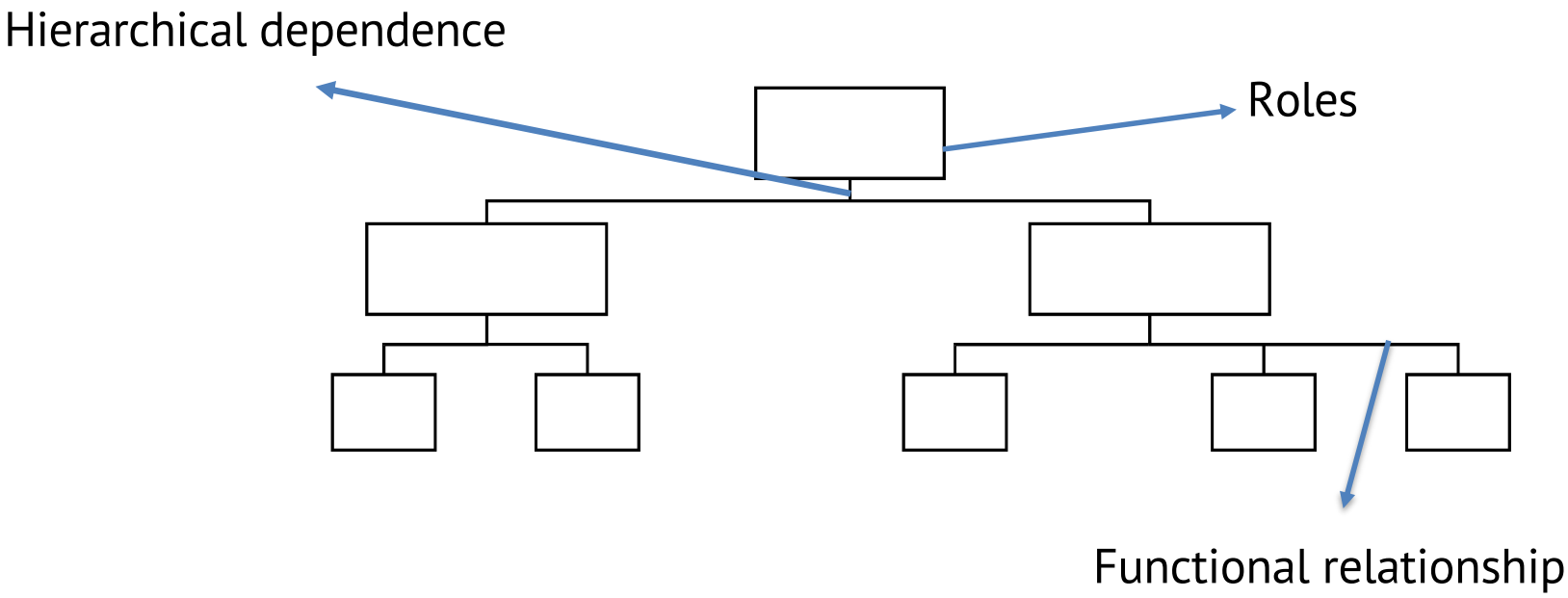
**As the number of members and the complexity of the organization grows, mutual adaptation and direct supervision are no longer sufficient; standardization is needed for coordination:**

3. **on processes**
4. **on results**
5. **on competences**

# Organizational chart

**Macrostructure**

An organizational chart is a graphic representation that takes the form of an inverted tree in which the relationships of dependence, or in any case of interconnection, between single individuals or subsets of the organization (organizational units) are represented.



# Organizational chart

**Macrostructure**

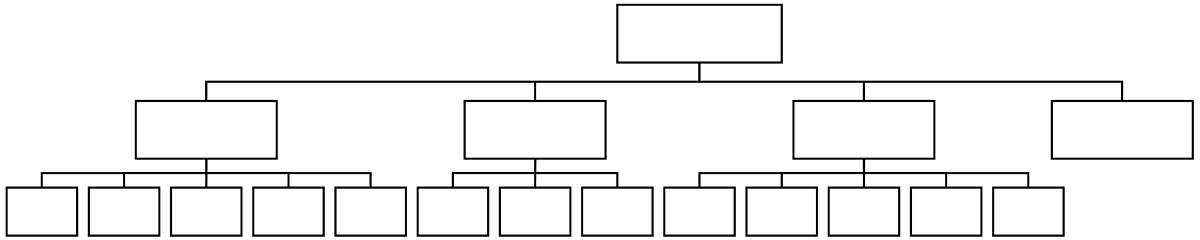
- **Top management**: those who run the company (management unit) and are responsible for strategic decisions
  - Led by the head of the organization (CEO, General Manager, President)
- **Span of control**: number of units reporting to/accounting to a single unit → Horizontal dimension of the organizational chart
- **Depth of control**: number of hierarchical levels in the organizational chart → Vertical dimension of the organizational chart

# Organizational chart



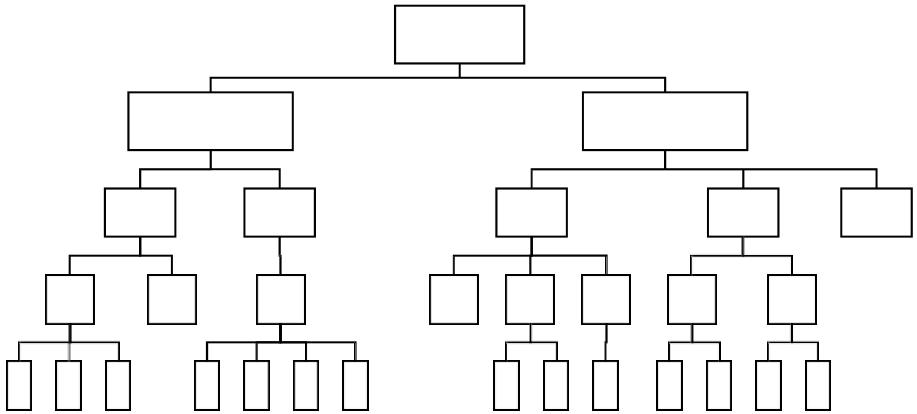
**Examples of horizontal organizational chart**

- **Breadth** of control (first line): 4, **Depth** of control: 3



**Examples of vertical organizational chart**

- **Breadth** of control (first line): 2, **Depth** of control: 5



**Macrostructure**

The aggregation of multiple roles (and tasks) within the same operational unit can obey different criteria:

- **Numerical grouping:** Organizational units are formed by dividing individuals into groups of adequate size to perform the task and be managed by a single manager.

- **Input-based grouping:** Tasks that use common resources (technology, equipment, skills) are grouped together.
By function
By skills

- **Output-based grouping:**
By product
By market
By customer

**Macro-structure**

The result of applying grouping criteria (at different hierarchical levels) is the so-called organizational structure.

Organizational structures are the result of choices regarding the division of labor and coordination guided by specific criteria.

In practice, five types of organizational structures can be identified:

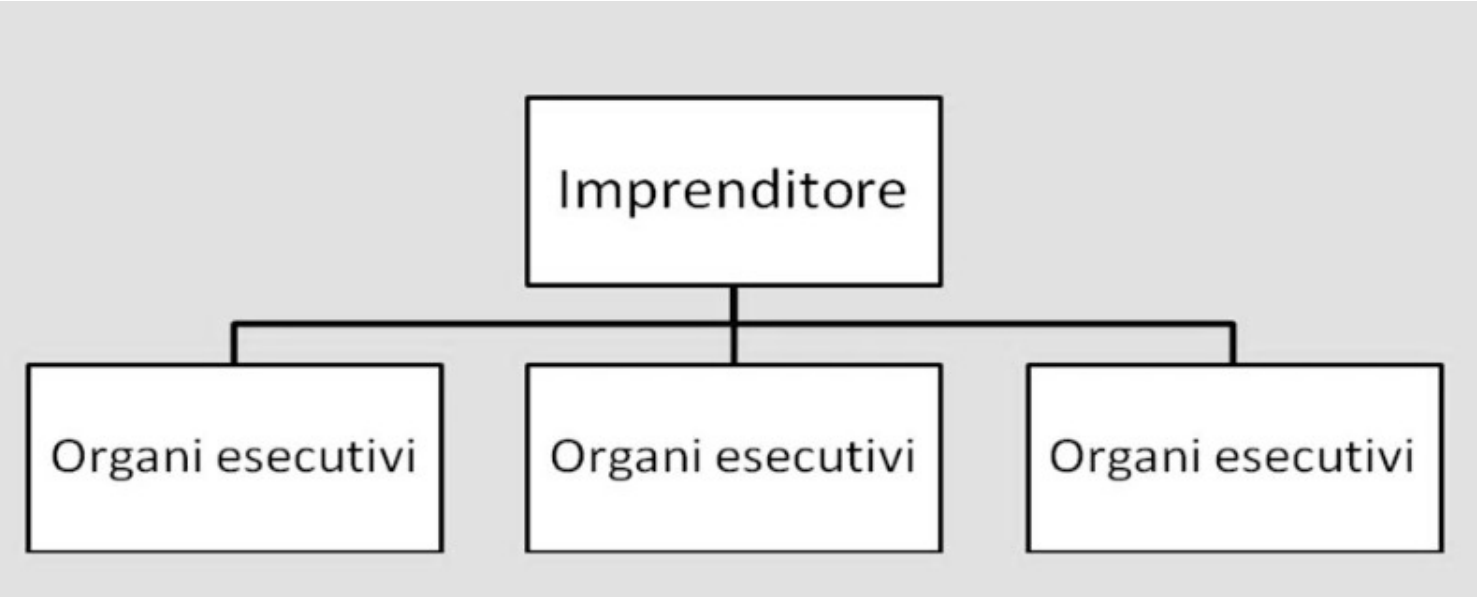
- **Simple structure**
- **Functional structure**
- Divisional structure: geographical area/product categories
- **Matrix structure**

Real organizational structures are often not pure, but include different types of structure at different hierarchical levels. For example, a functional structure at the first level and a divisional structure by product at the second level.

# Simple structure

**MACROSTRUTTRA**

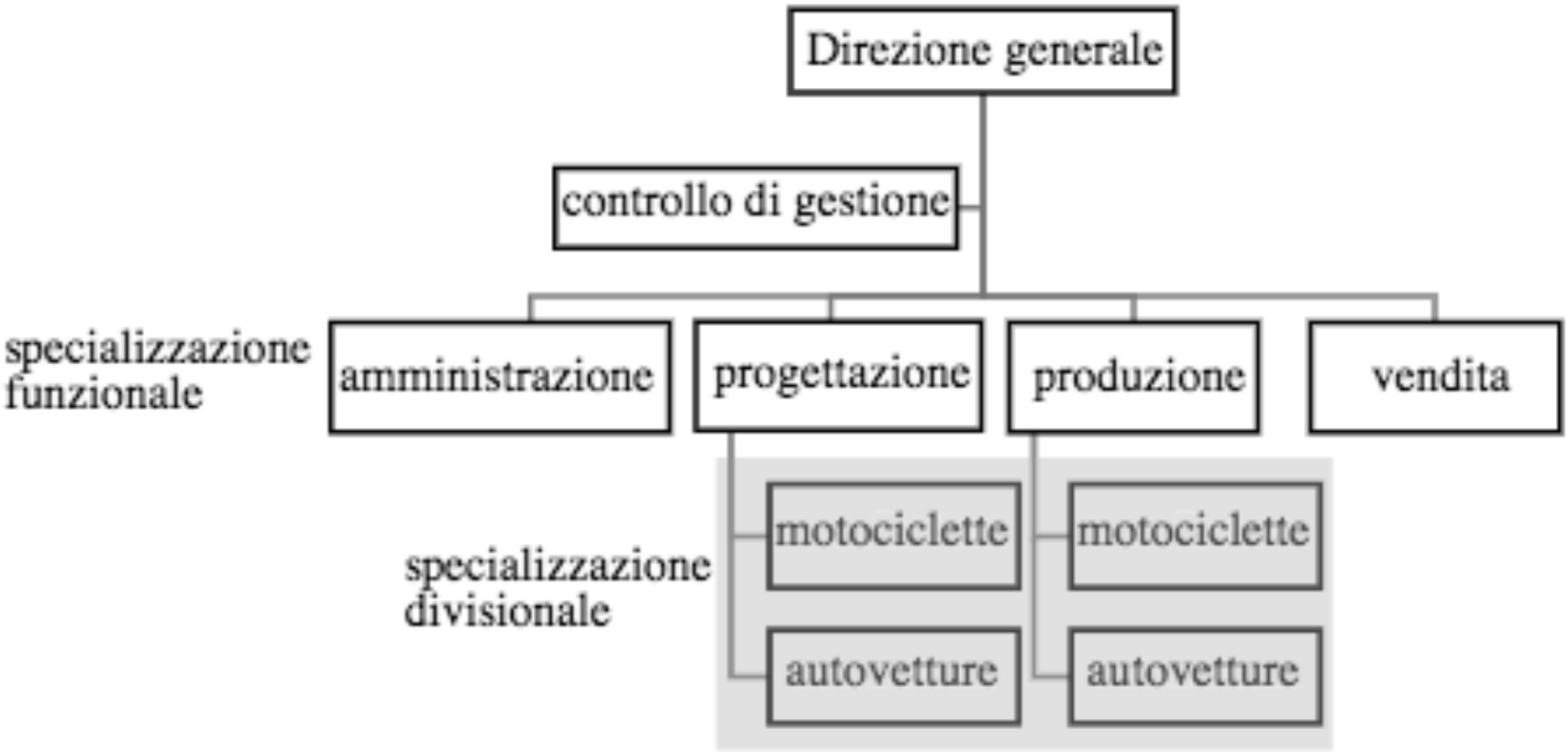
## Organizational chart



# Functional structure

**MACROSTRUTTRA**

## Organizational chart



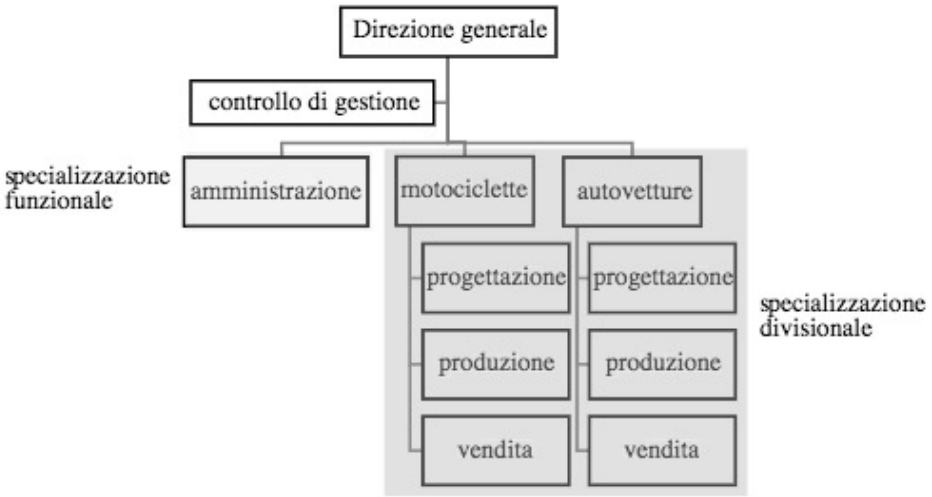
# Divisional structure

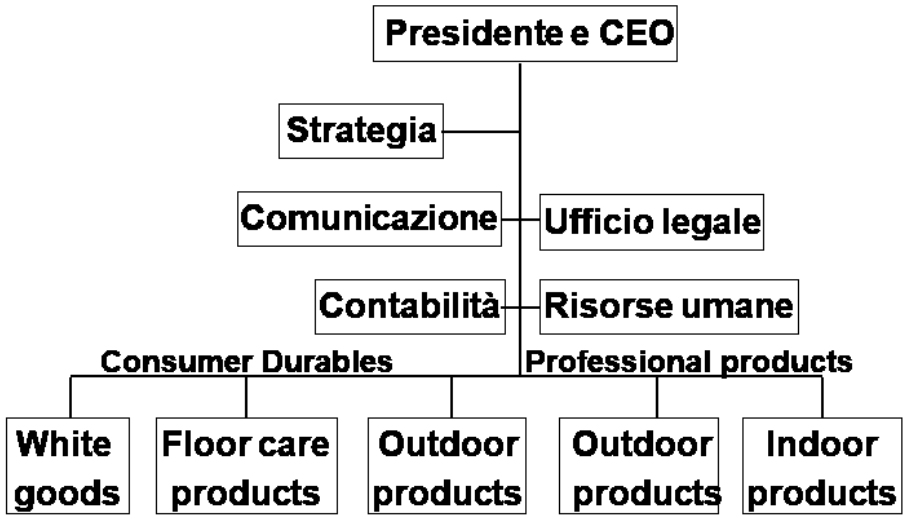
**Macrostructure**

## Organizational chart

**Per product**: division and specialization based on products

**Per client**: division and specialization based on markets served

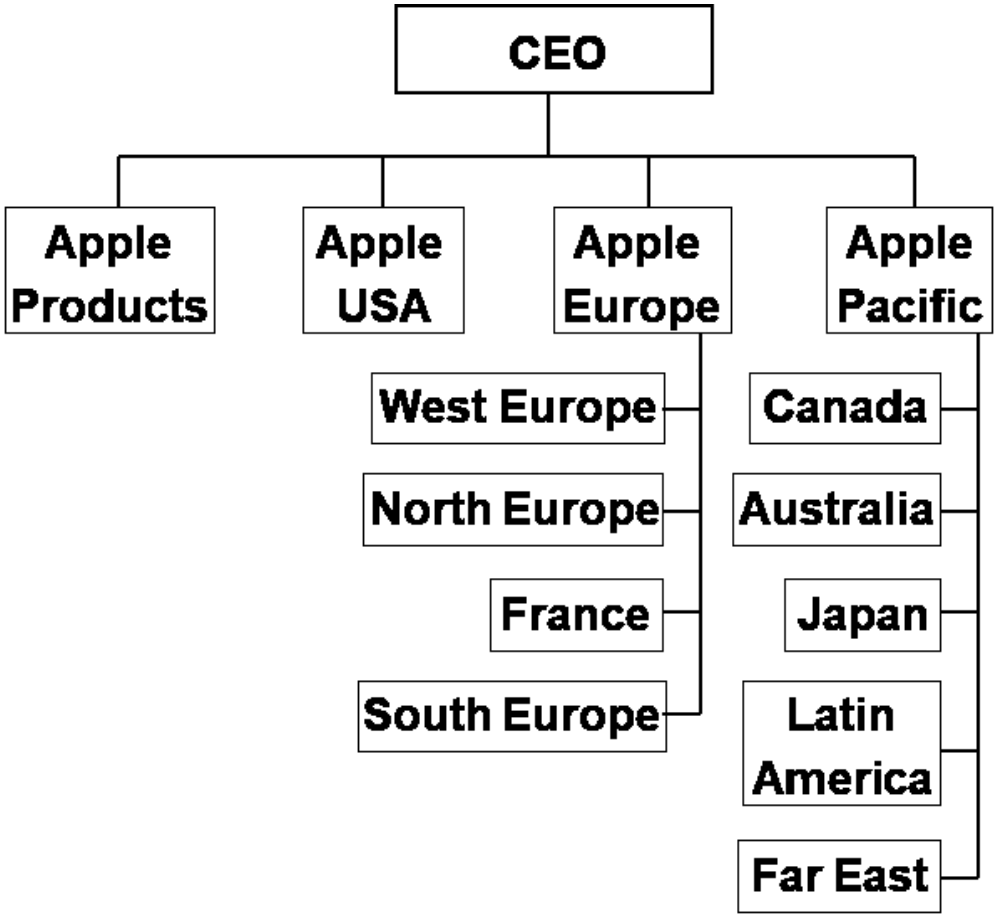


# Divisional structure

Macrostructure

## Organizational chart

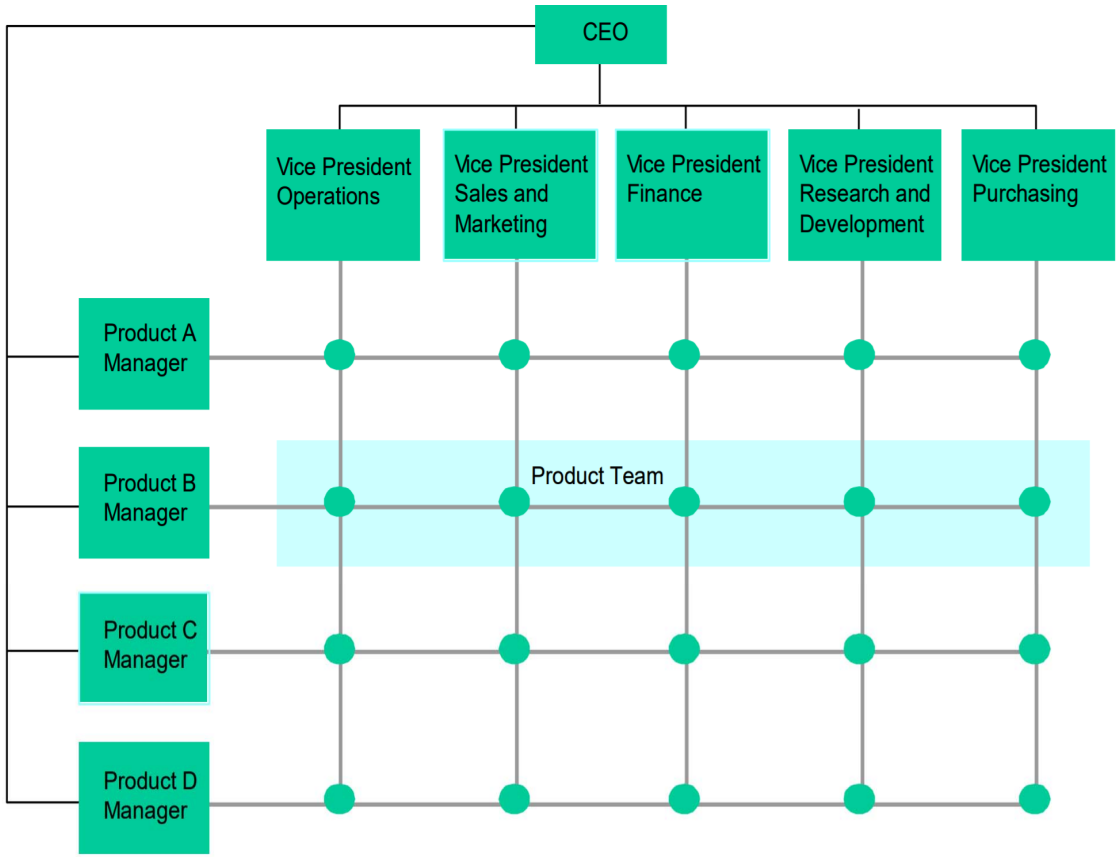



**By geographic area:** Divisions, which are also called branches or subsidiaries, are created based on geographic areas

# Matrix structure

Macrostructure

## Organizational chart



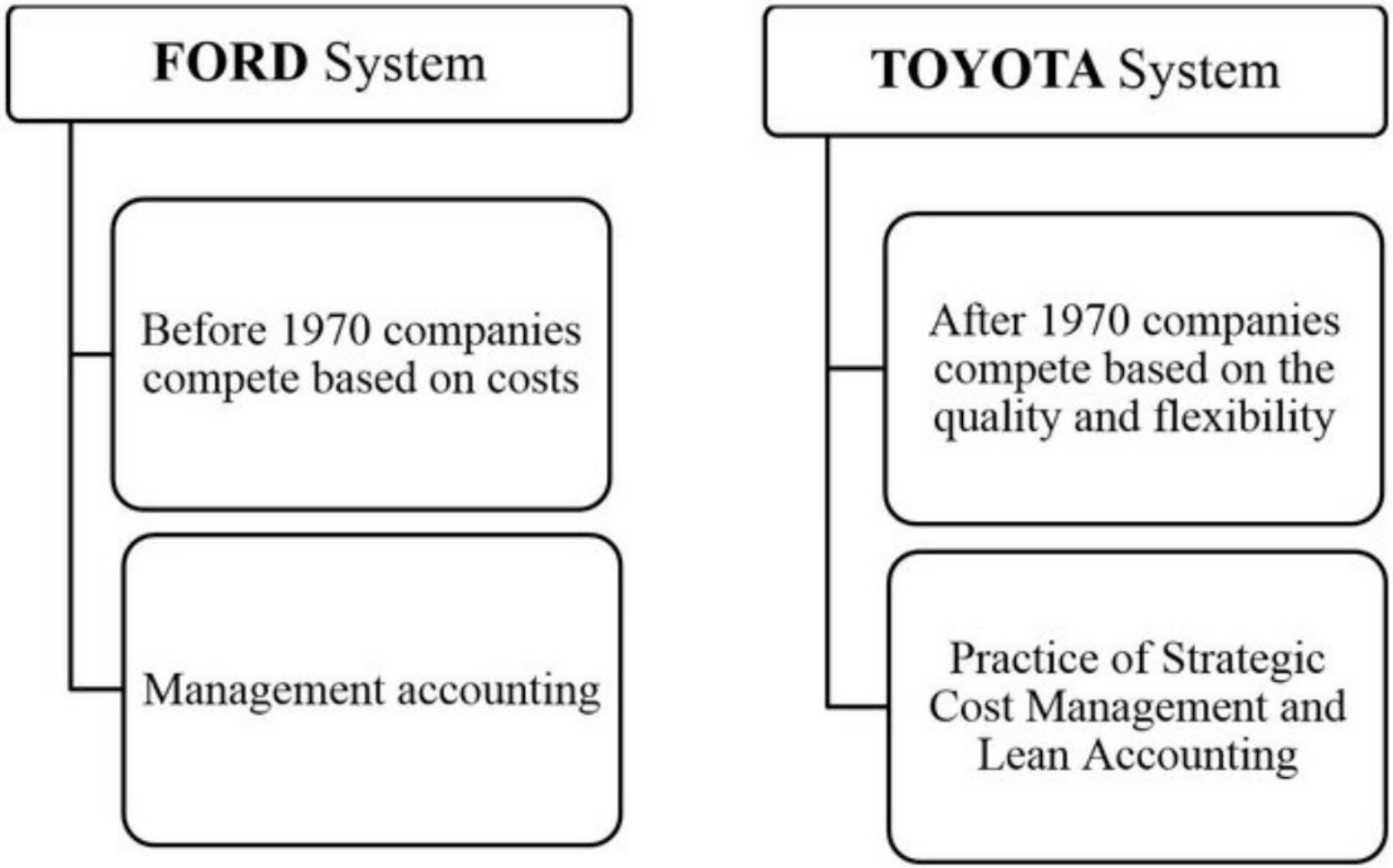
# The organization today

Evolution from post-fordism (1980-2000)

**Lean Production:** Krafciz, 1988; Womack, Jones, Ross 1988 (Massachussets Institute of Technology (MIT) »

- Triumph of the lean production system" Case Study Toyota
- From quantity to quality
- No waste, minimize storage
- Flexible, incremental and scalable approach
- Decentralization, importance of the supplier network
- Multitasking and non-rigid and divided skills

# The organization today



# The organization today

Evolution of the concept of business and organization at a systemic level (2000-2010)

**New Capitalism and high Digitalization:**

- Intense use of ICT Logistics
- Industrial districts (Automotive in Emilia; Paper in Lucca; Sylicon Valley)
- Entrepreneurial ecosystems

# The organization today

Evolution of the concept of business and organization at a systemic level (2010-today)

**Advanced Digitalization:**

- Use of algorithmic production mechanisms (3D printers)
- Overcoming the concept of mass production vs. specialized production thanks to artificial intelligence
- Global digital supply chain
- Role of labor:
  - Continuous and perpetual human-digital machine interaction
  - Algorithmic performance control

# Control the trade off between Technology and Ethics is crucial for the future of organizations

Business evolution faces this dilemma, which involves entrepreneurs, and policy-makers

